

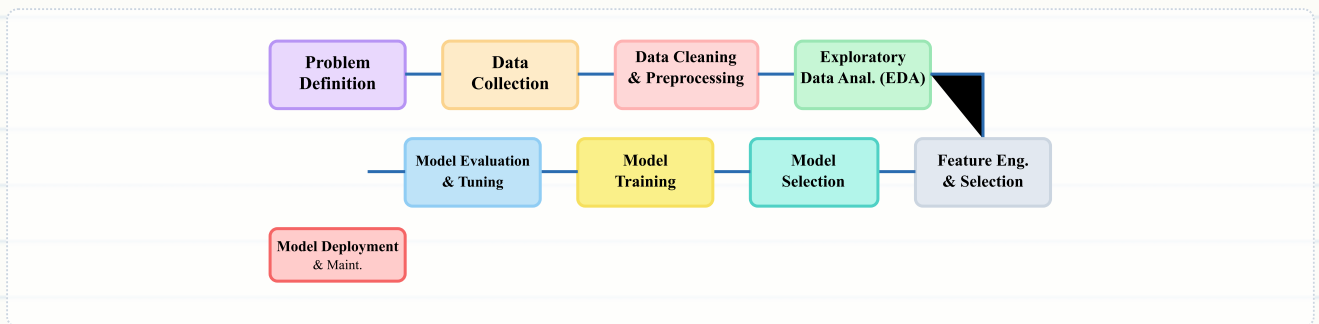
# Introduction to EDA & Feature Engineering

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Exploratory Data Analysis (EDA) and Feature Engineering are two of the most important steps in the machine learning workflow.

They help transform raw data into meaningful insights and build a strong foundation for accurate and robust models.

## Machine Learning Lifecycle



## Exploratory Data Analysis (EDA)

EDA is the process of exploring, understanding, and summarizing datasets before applying any machine learning algorithms.

It helps in:

- Identifying data patterns and trends
- Detecting outliers and missing values
- Understanding variable distributions
- Building intuition for model selection

## Common techniques in EDA:

- Descriptive statistics ( mean, median, mode, variance , etc.)
- Data visualization (histograms, scatter plots, box plots, heatmaps)
- Correlation and covariance analysis
- Handling missing values and outliers

## Feature Engineering

Feature Engineering is the process of creating, transforming, and selecting features to improve model performance. It bridges the gap between raw data and machine learning algorithms.

### Key steps in Feature Engineering:

- Handling categorical variables ( encoding techniques )
- Feature scaling ( normalization & standardization )
- Creating new features from existing ones
- Feature selection and dimensionality reduction (PCA, Lasso, etc.)
- Domain knowledge-driven transformations

**Mistake:** Don't apply feature scaling before splitting your train and test sets to prevent data leakage!

## Why are EDA & Feature Engineering important?

- Improve model accuracy and reliability
- Reduce noise and redundancy in data
- Enhance interpretability of models
- Save time and resources during training